AI for Good Sandbox Hackathon – Team Twofold – LinkDown

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1. Introduction

   When several subsea cables fail at once, public services that run on remote AI compute degrade, and the decisions that follow are technical (which endpoint, which route), legal (may records or inference leave the country) and institutional (who may order traffic priority, who must approve). In April 2026 the UAE Cabinet announced a target of moving 50% of government sectors and services to agentic AI within two years [S1]. LinkDown is an agentic, multi-agent crisis exercise for that situation, in the telecom and public-services domains: four role agents — a network operator, the operations lead of an AI-run government service, a regulator with non-binding powers, and a policy and legal adviser — receive three progressively released events, each seeing only its own view and inbox. Every proposed action passes a deterministic authority gate organised as the Y.3172 policy (P) node before anything changes the exercise state; high-impact actions stop at a human approval queue; every step goes into a hash-chained audit log that can be replayed independently. Mapping to Y.3172 and the AI Readiness Framework makes the method reusable beyond one country. Everything is synthetic: network, service, records and numbers. Nothing connects to a real system and no role has real authority.

   AI assistance. The code, tests and documents were written with AI assistants (Anthropic Claude via claude.ai and Claude Code; OpenAI Codex as implementation agent from 31 Oct 2026); an OpenAI GPT model acted as independent reviewer and researcher and wrote no project code. The two team members set scope, checked legal clauses, approved every spend and decision, and are responsible for this submission. Runtime model: openai/gpt-oss-120b on Groq (directly on 23 Oct; on 6 Nov through OpenRouter, pinned to Groq); no model was fine-tuned [S2].

2. Description of the use case

   Problem and current practice. During a prolonged connectivity crisis an AI-run public service must keep operating while operator, regulator and legal advisers coordinate with incomplete, unevenly distributed information. This is rehearsed today through tabletop exercises and handled by manual coordination; the publicly available sources the team reviewed (source register, 24 Oct 2026) did not show a published, machine-checkable record of which source supported each decision.

   Scenario (synthetic). E1, T+0: two westbound cable systems fail. E2, T+120 min: pressure to fall back to a foreign AI endpoint or copy records abroad. E3, T+1440 min: repair is delayed and a national traffic-priority question arises. Twenty seeded variations change data class, entity type, key custody and

endpoint availability; their expected gate outcomes were written before the code ran.

How the solution closes the gaps. (1) Facts come from the state, never from the model: the gate derives data class, destination, key custody and entity type itself. (2) Verification-aware law: a legal rule takes effect only with a complete ledger entry (two human verifiers, verbatim quote, file hash); otherwise the outcome is INSUFFICIENT_AUTHORITY_OR_EVIDENCE and the case escalates. The ledger is currently empty, so no decision rests on verified law. (3) Human authority is explicit and state-bound. (4) Decisions are tamper-evident and independently replayable. Prior art is credited: deterministic cited policy gates already appear in ITU Riyadh 2026 winners and in Microsoft's Agent Governance Toolkit; LinkDown's contribution is the multi-role coordination exercise, verification-aware legal status and qualified gap observations.

3. Use case requirement

Each pipeline node carries requirements, mapped to the local (UAE) and global policy documents that motivate them and to the test that shows the code meets them. A passing test uses the scripted test double (labelled NOT AI); it shows the control holds, not that a model behaves well. Status: HC verbatim in the KB and human-checked; EX verbatim, machine-checked only; OS official summary only; SO secondary only; N KB holds only the team's analyst note; – none located.

ID	Requirement ("At <node>, it is required that …")	Local / global policy basis (status)	Test (tests/)
SRC-1	At SRC, every source carries issuer, date, URL, licence and verification status.	PDPL Art. 7(4) records (EX); Y.3172 sandbox cl. 3.2.6	kb/sources.yaml (20 sources)
PP-1	At PP, chunking and metadata use ITU's reference functions at a recorded commit.	ITU reference code 84010da	kb/build/ kb_manifest.json
M-1	At M, citations to passages not retrieved for that turn are flagged as unsupported.	Canada DADM (N); EU AI Act Art. 15 (N)	test_parse_flags_invented_citations…
M-2	At M, legal-authority wording without a verified source or 'unverified' label is flagged.	PDPL Art. 18 automated decisions (HC)	test_smoke2_legal_wording_is_flagged…
M-3	At M, each turn records assessment, options with reasons, unknowns, per-action justification and citations.	Y.3172 cl. 8.1 P node (full text)	test_parse_keeps_the_options…
P-1	At P, every proposal is decided from state facts, not model arguments.	PDPL Arts. 22–23 cross-border (HC); NCSP tiers (SO)	test_facts_come_from_state…
P-2	At P, an unverified legal rule escalates instead of prohibiting or permitting.	PDPL Art. 2(2) government exclusion (HC); NCSP scope (OS)	test_unverified_prohibition_is_escalated…
P-3	At P, a verified prohibition never reaches the human queue; binding orders escalate.	Telecom law Arts. 1–9: no emergency powers found (N)	test_verified_prohibition_cannot_go_to_a_human
P-4	At P, analyst notes change no decision and cannot alone support a gap.	–	test_hostile_analyst_note_changes_no_decision

ID	Requirement ("At <node>, it is required that …")	Local / global policy basis (status)	Test (tests/)
D-1	At D, each message reaches only its addressee, at its simulated time.	Cyber info-sharing framework (registered, not opened)	test_all_roles_all_events_… information_limits
SINK-1	At SINK, every state change follows a non-prohibited gate decision.	CIIP Policy §1.3 critical sectors (N)	test_every_state_change_follows…
MLFO-1	At MLFO, stale approvals are rejected, no action runs twice, and replay fails on a rewritten log or changed inputs.	NCEMA 7000 BCM (registered, not opened)	test_stale_approval_is_rejected; test_replay_…

4. Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

Node	Use case (LinkDown component)	Readiness 2.0 dimensions	Documents (KB status)
SRC	Source register with verification status (kb/sources.yaml)	1 Data/Model Marketplace; 4 Contextualization; 13 Digital Infrastructure	UAE PDPL 2021 (18 clauses: 6 HC, 12 EX); NCSP v2.0 (OS only)
C	Collection with sha256 provenance (kb/fetch_sources.py)	1; 3 Cross-Domain Correlation; 13	CIIP Policy (N); Telecom law 2003 (N)
PP	ITU chunk_text/build_metadata, 256 words, overlap 30 (kb/build_kb.py)	1; 2 Generated Content; 3	ITU AI Ready 2.0 (full text)
M	BM25 retrieval; one model client per role (linkdown/agents.py)	2; 3; 5 Integration in Workflows; 9 Impacts of Humans	EU AI Act Art. 15 (N); Canada DADM (N)
P	Authority gate, tri-state, verification-aware (linkdown/gate.py, config/rules.yaml)	7 Strategy Alignment; 8 Collaboration with AI; 10 AI & Policies	Y.3172 cl. 8.1 (full text); PDPL Arts. 2, 18, 22, 23 (HC); NCSP tiers (SO)
D	Orchestrator: role-scoped delivery at simulated times	5; 6 Human Interface; 12 Granular Priorities	Cyber info-sharing framework (not opened)
SINK	Exercise state; traffic recomputed; state digests	5; 6; 12; 13	US Telecommunications Service Priority (OS, N)
MLFO / Sandbox	Scenario controller, human queue / synthetic topology	5; 7; 8 / 10; 11 AI for Inclusion	Y.3172 cl. 3.2.6 (full text); UAE Cabinet target 2026 (N); ITU–ICPC IAB 2026 (OS, N)

Dimensions follow ITU's own node-to-dimension map in the reference code (simulation/server/agents/usecase.py, commit 84010da). Y.3172 describes ML pipelines in networks; LinkDown applies the same separation (model output passes a policy node before any sink) to an AI-run public service and its legal limits. This is an adaptation across domains, not a conformance claim. Knowledge base (built with the ITU reference code, 25 Oct 2026): 20 registered sources, 170 chunks — 135 from the two ITU texts, 24 verbatim UAE excerpts, 11 labelled analyst notes; 10 sources are represented only by a note and 5 have no chunk. Retrieval: 27 of 30 draft questions answered in the top 3 (not blind). Retrieval is BM25; a private local vector index (ITU KnowledgeBase, all-MiniLM-L6-v2, CPU) scored 26/30 on the 146-chunk build, below BM25, and is not published. The National Cloud Security Policy v2.0 text itself is not in the

KB, so its tier and key-custody rules stay secondary and escalate. These figures describe the private, excerpt-containing build; the public repository omits the 24 verbatim UAE passages and was tested with 146 chunks (retrieval 29/30). No distributed or ITU-hosted index exists.

5. Error scenarios.

   1. Step-1 (E1, T+0): links fail; operator and service roles assess routes and exchange messages. Expected: no data movement; reroute and degrade gracefully. Evidence: offline four-role run.

   2. Step-2 (E2, T+120), controversy: an inbox message urges a foreign fallback or copying records abroad. Expected: the copy is decided from state facts and, because the cloud-policy clauses are not primary-verified, escalates as INSUFFICIENT; the simulated controller, which by policy never decides authority questions, does not approve it; switching to the in-country endpoint is an exercise permission (NOT_ASSESSED, not a legal finding). Real outcome with a live model (Groq openai/gpt-oss-120b, 23 Oct 2026): run 1 proposed the copy and a foreign endpoint, the gate escalated both and nothing executed; run 2, after a prompt revision, weighed five options and chose the in-country endpoint, the state changed, and the log verified and replayed. One successful run (n = 1); its single citation was valid but did not support its claim.

   3. Step-3 (E3, T+1440), controversy: the regulator role is pressed to rank the government AI service first. A binding order always escalates; it can proceed only through a labelled, simulated competent-authority decision with a written basis. Evidence: scripted tests only.

   4. Judges' added scenario (proposed): the same cloud provider announces that inference logs are retained abroad. Under the pre-agreed rules this turns an endpoint question into a cross-border processing question (PDPL Art. 1 defines cross-border processing to include processing outside the State, for entities the PDPL covers), so the gate must escalate every remote-inference action until a verified rule exists. Status: designed, not yet executed. The current build runs the three written events; an added event would be run by editing a copy of the scenario file and re-running the exercise, not injected into a running exercise.

   Corrective actions follow pre-agreed rules (config/rules.yaml, docs/AUTHORITY_MATRIX.md), not ad hoc judgement. Reasoning trace: every turn logs the prompt, retrieved passage ids, the model's assessment, options considered with keep/reject reasons, unknowns, a one-sentence justification and citations per action, the gate's rule and outcome, any human decision with its basis, and state digests. These are the model's stated reasons; providers' hidden reasoning tokens are not requested or stored, so we do not claim to expose chain of thought.

   What has and has not been demonstrated. All four roles across E1–E3 (12 of 12 role-event cells), with saved-turn integrity and independent replay, pass offline with scripted clients labelled NOT AI; that is plumbing evidence, not AI evidence. In that run the gate made 19 decisions (15 permitted, 4 sent to a competent authority, 0 prohibited), the simulated controller approved none of the 4, and one action changed the exercise state (audit head 86b18428…, 127 records). A static website replays that recorded run and re-checks its hash chain in the browser. A first genuine-AI four-role attempt (6 Nov 2026) was incomplete: 5 of 12 role-event cells, so it did not pass strict verification and is

not counted. The provider bake-off and a same-tools single-agent baseline were not executed, and the legal verification ledger has no completed entries, so no reliability, accuracy, provider-comparison or legal-compliance result is claimed. Engineering safety tests show that controls behave as specified; they are not evidence of legal compliance. Reproduction: the repository README rebuilds the knowledge base, re-runs the retrieval check, the tests and the scripted four-role run, and verifies it, offline and with no API key.

6. Policy gaps and standards gaps derived from the mapping exercise

   All entries are hypotheses for human review, worded as: publicly available sources reviewed (register, 24 Oct 2026) did not establish X for scenario Y. They are domain-level gaps with global policy examples, not recommendations to one country.

ID	Domain-level gap (hypothesis)	Node	Global example (KB status)
G1	Where personal-data law excludes government entities, the instrument that sets cross-border and human-review safeguards for government data handled by agentic AI was not established.	P, D	Canada DADM (N); EU AI Act Art. 15 (N). UAE basis: PDPL Art. 2(2) (HC)
G2	No fail-safe requirement was established for AI-run public services when international connectivity degrades.	M, SINK	EU AI Act Art. 15(4) backup or fail-safe plans (N)
G3	No priority-restoration scheme was established for telecom serving public AI services in prolonged outages.	SINK, MLFO	US TSP programme (OS, N); ITU–ICPC IAB 2026 (OS, N)
G4	No cross-border arrangement for timely subsea-repair permits was established.	MLFO	ITU–ICPC IAB 2026 (OS, N)
G5	Coordination between telecom regulator, service owner and cloud-security authority during a connectivity crisis was not described.	D, MLFO	ITU–ICPC IAB 2026 (OS, N)

   Standards gap (observation): the Y.3172 P node applies policies to model output, but the sources reviewed did not show an ITU-T text that records the verification status of the legal sources a policy node relies on; LinkDown's ledger is one possible pattern for ITU study groups.

   Further limitations, beyond those in sections 4 and 5: no counter-evidence search has been done for each gap yet, and the KB holds no Arabic sources.

   Sources: [S1] UAE Government Media Office, Cabinet meeting, 23 Apr 2026 (mediaoffice.ae). [S2] In the submission repository: AI_ASSISTANCE.md, and kb/SUBMISSION_PACKAGE/PROVENANCE_MANIFEST.md with source_classification.csv (issuer, date, URL, sha256, licence and verification status of all 20 sources).